

Even Realities G1 BLE Protocol

This doc aims to outline the BLE protocol for the Even Realities G1. The G1 operates with two BLE radios, one for the left arm and one for the right. Some commands need to be sent individually to both arms and some are only sent to one arm. Each radio operates as a stream of packets using the Nordic BLE UART.

Generic Command Response

Many commands will simply respond with a generic success/failure message. The header will repeat the command id back, and one additional byte which will be either success or failure. This is recorded here and referenced by all commands that operate this way to avoid duplicating the same response for all such messages.

Response

Command	Subcommand	Description
XX	C9	Success
XX	CA	Failure
XX	CB	Continue data

Commands

Commands are messages the App can send to the glasses. The glasses may optionally respond.

Silent Mode Control (0x03)

This will activate or deactivate the silent mode of the glasses. Send to **both** arms. [Response is generic.](#)

Command	Subcommand	Description
03	0C	Silent Mode On
03	0A	Silent Mode Off

Notification Configuration (0x04)

Send over the config for notifications. This has JSON with some booleans and an allowlist for which apps to display notifications for. Send to **left** arm. [Response is generic](#).

Command	Chunk Count	Sequence	Payload (JSON) (Max size 180 bytes)
04	01~FF?	00~FF	<pre>{ "calendar_enable":true, "Call_enable":true, "Msg_enable":true, "Ios_mail_enable":true, "app":{ "List":[{ "id":"com.app", "name":"App Name"}, ...], "enable":true } }</pre>

Set Dashboard Mode (0x06 07)

Set the dashboard mode, currently has three modes: minimal, dual, full

Command	Subcommand	Sequence	Api ?	Main Mode	Sub Mode	Description
06	07	00 00-FF FF	06	00	00 - 06	Full
06	07	00 00-FF FF	06	01	00 - 06	Dual
06	07	00 00-FF FF	06	02	00	Minimal

Sub Modes

Main Mode	Description
00	Notes
01	Stock (graph)
02	News
03	Calendar
04	Navigation
05 +	Empty

Response

The response repeats back the sequence

Command	Subcommand	Sequence	??	Mode
06	07	00 00-FF FF	06	00

Dashboard ?? (0x06 0C)

Command	Subcommand	Sequence	
06	0C	00 00-FF FF	

Set Dashboard Time and Weather? (0x06 15)

Set Time (maybe weather too)??

Command	Subcommand	Sequence	Api ?	Epoch Time 32bit (s)	Epoch Time 64bit (ms)	Weather	Temp C	C/F	24H/12H
06	15	00 00-FF FF	01	8C 45 A8 67	e5 ad 4f e9 94 01 00 00	01-10	04	00/01	00/01

Weather Icons:

Weather Code	Description
00	None
01	Night
02	Clouds
03	Drizzle
04	Heavy Drizzle
05	Rain
06	Heavy Rain
07	Thunder
08	Thunder Storm
09	Snow
0A	Mist
0B	Fog

0C	Sand
0D	Squalls
0E	Tornado
0F	Freezing
10	Sunny

Response

The response repeats back the sequence number

Command	Subcommand	Sequence	Payload
06	15	00 00-FF FF	00 01

Microphone Control (0x0E)

enable:

0: Disable the MIC (turn off sound pickup).

1: Enable the MIC (turn on sound pickup).

Response from Glasses

Command: 0x0E

rsp_status (Response Status):

0xC9: Success

0xCA: Failure

enable:

0: MIC disabled.

1: MIC enabled.

Set Quick Note (0x1E)

TODO

Fetch FW Info (0x23 74)

Send and the device should respond with fw build information

Command	Subcommand
23	74

Response

This response kind of sucks because it has no header and is just raw ASCII data. Currently it always starts with “net”

Hex Payload (203 bytes)	ASCII Payload
020320cf00cb0004001b12006e6574206 275696c642074696d653a20323032342d 31322d32382032303a32313a35372c206 17070206275696c642074696d65203230 32342d31322d32382032303a32303a343 52c2076657220312e342e352c204a4244 20446576696365494420343031300a0d0 00000000000000000000000000000000 00000000000000000000000000000000 00000000000000000000000000000000 00000000000000000000000000000000 00000000000000000000000000000000 00000000000000000000000000000000 00000000000000000000000000000000	net build time: 2024-12-28 20:21:57, app build time 2024-12-28 20:20:45, ver 1.4.5, JBD DeviceID 4010

Heartbeat (0x25)

Needs to be sent periodically to keep the device connected. Disconnection happens after 32 seconds, so send it every 30 seconds.

Command	Subcommand
2C	01

Fetch Battery Info (0x2C 01)

Send and the device should respond with a battery info packet

Command	Subcommand
2C	01

Response

Header	?	Battery Percentage	?
2C	66 ?	00~64	In case, arms closed, charging. L 00 e6 (e7) 5d 24 00 00 00 01 04 05 0000000000000000 R 64 ef(f0) 80 19(18) 01 04 05 01 04 05

			0000000000000000
--	--	--	------------------

Time Since Boot (0x37)

Fetches the time since boot from the device in seconds..

Command
37

Response

Header	Payload
37	49 1a 00 00 01 00 00 00 00 00 00 00 00 00 00 00 00 00

?? (0x3D)

Get Buried Point Data (0x3E)

Fetch buried point data, which is essentially user usage tracking:

<https://www.php.cn/faq/446290.html>

Command
3E

Response

Header	Success	Payload (197 bytes ?)
3E	C9	R Example censored as it may contain private data.

Display Notification (0x4B)

Forward a notification to the glasses. Send to **left** arm. [Response is generic.](#)

Command	Chunk Count	Sequence	Payload (JSON) (Max size 180 bytes)
4B	01~FF?	00~FF	

Init?? (0x4D)

Not sure what this does, sent to the left only. [Response is generic.](#)

Command	Payload?
4D	FB?

Retrieve Audio Recording (0xF1)

- Command: 0xF1
- seq (Sequence Number): 0~255
- data (Audio Data): Actual MIC audio data being transmitted.

Field Descriptions

- seq (Sequence Number):
 - Range: 0~255
 - Description: This is the sequence number of the current data packet. It helps to ensure the order of the audio data being received.
- data (Audio Data):
 - Description: The actual audio data captured by the MIC, transmitted in chunks according to the sequence.

Example

- Command: 0xF1, with seq = 10 and data = [Audio Data]
- Description: This command transmits a chunk of audio data from the glasses' MIC, with a sequence number of 10 to maintain packet order.

Messages

These are messages the glasses will send to the app unprompted.

State Change (0xF5)

Touchpad events on FW v1.4.5 and below will also initiate events on the glasses. In FW v1.X.X, these events were made remappable, so the on glasses functionality is not mentioned here since it can be changed by the user.

Header	Sub	Payload	Description
--------	-----	---------	-------------

F5	00	??	TouchPad Double Tap
F5	01		TouchPad Single Tap
F5	04		TouchPad Triple Tap
F5	05		TouchPad Triple Tap
F5			
F5	0F	00~64 (00 x 21)	Case Battery percent
F5	11		BLE Paired Success
F5	17		TouchPad pressed and held
F5	18		TouchPad pressed and released
F5			
F5			
F5			
F5			
F5			